

KAHBOD AEINI

Calgary, AB

Relocating to Waterloo, Jan 2026

kahbod.aeini@ucalgary.ca — LinkedIn — Website

SUMMARY

Versatile software and data engineer with strong foundations in Python, C/C++, and production data systems (PostgreSQL, MongoDB, Docker, Linux). Experience designing and benchmarking concurrent algorithms, building robust ETL and ML pipelines, and delivering reliable data infrastructure that improved throughput and reduced incidents. Proven track record translating research prototypes into production-ready code, leading small teams, and automating analytics and reporting for reproducible results. Seeking software or data engineering roles focused on scalable backend systems, data platforms, and ML-enabled services; relocating to Waterloo Jan 2026.

SKILLS

Languages: Python, C/C++, Java, R, Haskell, Prolog

Systems & Data: PostgreSQL, MongoDB, MySQL, ETL, Docker, Linux, NixOS

ML / Tools: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, OpenCV

EXPERIENCE

Research Assistant — University of Calgary

Jun 2024 – Present

- Designed and analyzed randomized wait-free lock algorithms, proving space- and time-complexity bounds; results guided concurrent data-structure design for high-throughput systems.
- Implemented prototypes and benchmarks to measure practical performance under contention; optimized implementations reduced observed latency in microbenchmarks.

Research Assistant (Remote) — Johns Hopkins University

Feb 2023 – Aug 2023

- Built ML pipelines (Python/R) and feature selection workflows to predict immune cell composition from transcriptomic data; improved model AUC and reproducibility via robust preprocessing and cross-validation.
- Automated analysis and reporting, enabling faster iteration for collaborators and reproducible research artifacts.

Data Engineer — AI Lida

Feb 2022 – Sep 2022

- Designed and maintained ETL pipelines using Python and PostgreSQL; improved data reliability and throughput for analytics workloads.
- Led hiring and onboarding for 5 interns; introduced testing and monitoring that reduced production incidents.

SELECTED PROJECTS

Fraud Detection System (BSc Thesis)

GitHub

Developed and compared multiple ML models for sequence-based fraud detection; engineered feature pipelines for production inference.

Code Understanding / Prediction

KGTorrent (research)

Built deep-learning models and feature extractors for Python notebooks to support code-prediction and program understanding tasks.

Face Clustering with GCNs

IPM research

Developed graph-convolutional approaches for face clustering and linkage prediction; implemented evaluation pipeline and benchmarks.

Secure Chat App. Developed a secure chat app in Python in a team of three, secured by different methods of computer security as the final project of Data and Network Security course. (Git - Jun 2023)

Cloud Drive Collaborated within a six-member team to design and implement a file-sharing service. Leveraged technologies such as Django, Docker, and Minio while adhering to Agile methodology for efficient development and delivery. (Git - Dec 2022)

Information Retrieval System. Designed and implemented a retrieval system for twitter interaction and health articles with GUI, Classification and Clustering tweets and articles. (Git - Jun 2022)

Natural Language Processing. Implemented NLP methods on a speech dataset and demonstrated the conclusions. (Git - Jul 2022)

Garage Services App. Developed an Android app for using services of a Garage (As part of the final project in mobile programming course). (Git - Jul 2021)

Navigation App. Developed a Navigation app in Android with full functionality and identical features as Google Maps. (Git - Feb 2021)

Cryptocurrency App. Developed an android app which represents any Cryptocurrency latest price and candles online. (Git - Jan 2021)

Online Shop. Developed an online shop with GUI and client-server architecture in Java language. (Git - Jun 2020)

EDUCATION

University of Waterloo

Jan 2026 – (incoming)

PhD, Electrical and Computer Engineering

University of Calgary

2023 – 2025

Master of Computer Science

Thesis: *Space-Optimal Randomized Wait-Free Lock*

Sharif University of Technology

2019 – 2023

BSc, Computer Engineering

Thesis: *Fraud Detection from Sequences of Bank Transactions Using Various Machine Learning Algorithms*

AWARDS

Departmental Research Award, University of Calgary (2023)

Top 100 National Universities Entrance Exam, Iran (2019)